

The Veneer of Search

How AI Interfaces Earn Trust They Haven't Earned

Audit synthesis

Spatharioti et al. (2025), *CHI '25* | Li & Aral (2025), *arXiv:2504.06435*
Sharma, Liao & Xiao (2024), *CHI '24* | Wardle, Urbani & Wang (2025), *JMIR* 27, e79961

1.6 min 47% 4.41/5

AI search cut task time in half.

It **halved accuracy** on a hard task.

Users rated the experience *best of all*.

Spatharioti et al. (2025), Exp. 1, $N=90$ MTurk; vs. traditional search 3.4 min, 93% accuracy, 3.10/5.

AI search wears four veneers — each builds unearned trust

Veneer	What it looks like	What it does	Source
Speed	“AI search is fast”	Hides errors users cannot catch	Spatharioti
References	“AI cites its sources”	Builds trust even when fake	Li & Aral
Search	“User issues queries”	Most queries are confirmatory	Sharma
Utility	“AI is useful”	Overrides users’ own distrust	Wardle

Veneer 1: speed that hides errors

Faster searches, worse decisions, higher satisfaction.

LLM search halved task time — and halved accuracy on the hard task

Outcome	Traditional search	LLM-based search
Median task duration	3.4 min	1.6 min ($p < .001$)
Easy-task accuracy	92.3%	95.3% (NS)
Hard-task accuracy	93%	47% ($p < .001$)
Number of queries	2.5	1.9 ($p = .02$)
Query complexity (1–5)	1.8	3.4 ($p < .001$)

On the hard task, 30/51 LLM users issued only *one* query; none re-queried after a wrong answer.

Spatharioti et al. (2025), Exp. 1, Figs. 2–3; 4Runner adversarial item.

Users prefer LLM search 4.41 vs. 3.10 — and cannot detect its errors

Self-reported measure	Traditional	LLM
Overall search experience (1–5)	3.10	4.41 ($p < .001$)
Perceived reliability (1–5)	~ 4	~ 4 (NS, $p = .91$)

The interface fix exists — but only if it is added

Confidence highlighting (red on low-probability tokens, Exp. 2) more than **doubled hard-task accuracy** from 26% to 58% ($p < .01$). The fix lives in the UI, not the model.

Spatharioti et al. (2025), Exp. 1 & 2, Figs. 4, 6.

Veneer 2: trust from interface, not content

The references work *whether or not they are real*.

Reference links increase trust — and hallucinated ones work just as well

Treatment	β on trust	β on share
References (vs. plain GenAI)	+0.091 ***	+0.121 ***
Hallucinated vs. valid references	−0.021 (NS)	−0.008 (NS)
Uncertainty highlighting	−0.157 ***	−0.179 ***
Negative social feedback	−0.213 ***	−0.286 ***
GenAI label vs. traditional (same content)	−0.043 *	−0.141 ***

The mere *presence* of citations creates trust, regardless of whether they resolve. Most vulnerable: lower education, less tech experience, infrequent GenAI users.

Li & Aral (2025), pre-registered RCT, $N=4,927$ Prolific, Figs. 3–4.

Veneer 3: confirmatory querying disguised as search

When the interface answers, the user stops looking.

Conversational search induces $10\times$ more confirmatory querying than web search

Measure	WebSearch	ConvSearch(Ref)
Confirmatory query rate	1.46%	15.0–16.2%
Confirmatory agreement (post-task)	0.80	1.79–1.89
Confirmatory trust	0.35	0.79–0.89

ANCOVA $F(2, 100) = 4.71$, $p = .01$ on querying; $F = 6.61$, $p = .002$ on agreement. Cohen's D up to **0.78**.

Inline references did not change this: ConvSearchRef behaved like ConvSearch.

Sharma, Liao & Xiao (2024), Study 1, $N = 115$ Prolific, Table 1.

Consonant LLMs polarize — but dissonant LLMs do **not** correct it

Measure	Consonant	Neutral	Dissonant
Confirmatory query rate	42.9%	15.6%	12.3%
Confirmatory arguments in essays	51.7%	34.8%	15.6%
Confirmatory agreement	2.44	1.84	1.51

The asymmetry is the result. Consonant LLMs amplify polarization with Cohen's D up to **1.19**; dissonant LLMs fail to depolarize on every measure (all $p > .05$ vs. Neutral). Designing “opposing-view” tools will not undo selective exposure.

Sharma, Liao & Xiao (2024), Study 2, $N = 223$, Table 3.

Veneer 4: utility that overrides distrust

Users adopt what they doubt — if it is convenient.

56% of users adopt AI for health information *despite* distrusting it

- 15/27 (56%) used AI-generated health info despite expressing skepticism, when it “helped power further investigation.”
- 14/27 (52%) already integrate ChatGPT into their health-search routines; 23/27 (85%) reported favorably on its formatting or efficiency.
- AI is used as a “starting point”: keyword generation, summarization, then verified on Google or with a doctor — *utility*, not credibility, drives integration.
- Only 5/27 (19%) understood that training data shapes LLM outputs.
- 2/3 expressed concern that ChatGPT does not cite sources — yet still used it.

Wardle, Urbani & Wang (2025), think-aloud protocol, $N=27$, JMIR 27:e79961.

A skeptic would say these are lab effects, not real-world behavior

The objection. Spatharioti uses MTurk ($N=90$, no demographics); Sharma uses Prolific ($N=115-223$, closed-world docs); Wardle is qualitative, $N=27$ from four East Coast cities. Only Li & Aral has scale, and even that is a controlled vignette.

The response.

- **The effect sizes are large.** Cohen's $D > 1$ in Sharma; a 46-point accuracy gap in Spatharioti; β 's significant at $p < .001$ in Li & Aral. Small samples find them anyway.
- **Four very different methods converge.** Between-subjects RCT, factorial RCT, large pre-registered RCT, qualitative think-aloud — all four point to interface-driven effects exceeding content-driven ones.
- **The lab is the world.** Li & Aral's exposure audit: **42%** of US Google searches now return AI Overviews; **56%** for general-knowledge queries. The vignettes describe what most users now see.

Restoring user judgment requires interface-level, not model-level, fixes

Veneer	Reform	Concrete evidence
Speed	Confidence highlighting on uncertain tokens	Spatharioti Exp. 2: 26%→58% accuracy
References	Verified-source badges; degrade UI on broken links	Li & Aral: refs trusted regardless of validity
Search	Default to neutral retrieval; flag opinion drift	Sharma: dissonant LLM does <i>not</i> fix it
Utility	AI literacy on training data; source labels	Wardle: only 19% understood data shapes outputs

Each fix is interface-side; none requires changing the underlying model.

The interface earned the trust.

The model **did not**.